

BOARDROOM-READY MATH · 2026

# Math Layouts

Seven variants under one `math` class. For quants, mathematicians, physicists, ML researchers, statisticians, and economists. KaTeX renders `$...$` inline and `$$...$$` display; the layouts surround the equations with the structure each audience expects.

SECTION 01

The hero equation · bare **math**  
resolves to feature.

## The closed-form estimator.

$$\hat{\beta} = (X^{\top} X)^{-1} X^{\top} y$$

WHERE

- $\hat{\beta}$  — OLS coefficient vector
- $X$  — design matrix,  $n \times p$
- $y$  — response vector, length  $n$
- $X^{\top} X$  — Gram matrix,  $p \times p$ , must be invertible

SECTION 02

# The derivation chain · **math** **derivation**

# Derivative of $f$ from first principles.

|                                                            |                                             |
|------------------------------------------------------------|---------------------------------------------|
| $f(x + h) = f(x) + f'(x) h + O(h^2)$                       | <i>Taylor expansion, <math>n = 2</math></i> |
| $f(x + h) - f(x) = f'(x) h + O(h^2)$                       | <i>subtract <math>f(x)</math></i>           |
| $\frac{f(x + h) - f(x)}{h} = f'(x) + O(h)$                 | <i>divide by <math>h \neq 0</math></i>      |
| $\lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = f'(x)$ | <b>take the limit</b>                       |

SECTION 03

Definition · theorem · proof · math  
theorem

# Intermediate Value Theorem.

**DEFINITION.** A function  $f : [a, b] \rightarrow \mathbb{R}$  is *continuous* on  $[a, b]$  if  $\lim_{x \rightarrow c} f(x) = f(c)$  for every  $c \in [a, b]$ .

**THEOREM.** Let  $f$  be continuous on  $[a, b]$  and let  $y$  lie strictly between  $f(a)$  and  $f(b)$ . Then there exists  $c \in (a, b)$  with  $f(c) = y$ .

**PROOF.** Set  $S = \{x \in [a, b] : f(x) < y\}$ .  $S$  is non-empty and bounded; let  $c = \sup S$ . Continuity at  $c$  forces  $f(c) = y$ .  $\square$

SECTION 04

# Side-by-side · math compare

## Frequentist vs Bayesian point estimate.

### FREQUENTIST

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$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(y \mid \theta)$$

Maximises the likelihood — no prior. Uncertainty quantified by the sampling distribution of  $\hat{\theta}$  across hypothetical repeats.

### BAYESIAN

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$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta \mid y)$$

Maximises the posterior — conditions on the prior  $p(\theta)$ . Uncertainty is the posterior itself, no repeated sampling required.

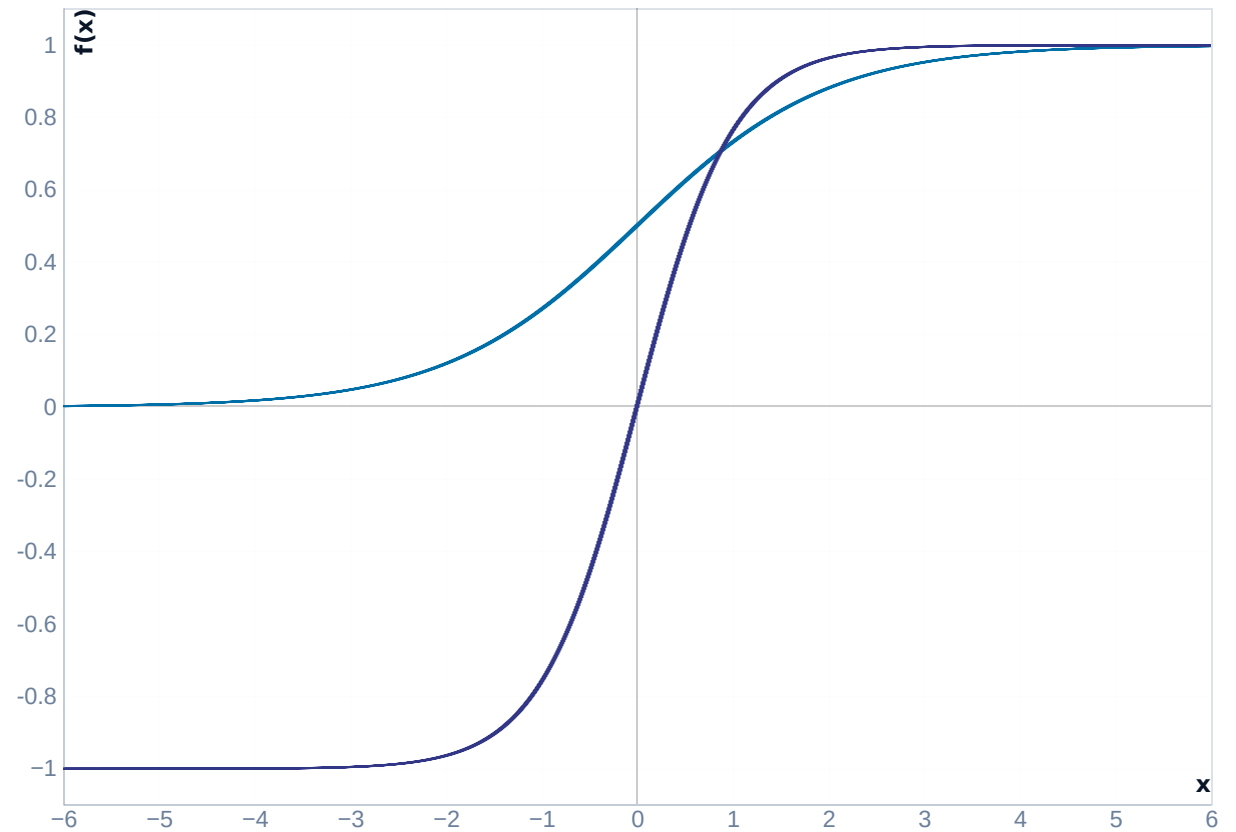
SECTION 05

# Equation + plot · math canvas

## The sigmoid.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Maps  $\mathbb{R} \rightarrow (0, 1)$ . *S*-shaped,  $\sigma(0) = 0.5$ , steepest slope at the origin.



SECTION 06

# Matrix · $\mathbf{matrix}$ and $\mathbf{matrix}$ $\mathbf{decompose}$

## The design matrix $X$ .

$$X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix}$$

- `SHAPE` —  $n \times (p + 1)$
- `ROWS` — observations
- `COLS` — intercept +  $p$  features
- `RANK` — full-rank for OLS to have a unique solution
- `COLUMN 0` — all-ones, absorbs the intercept

## Singular value decomposition.

$$A = U \Sigma V^{\top}$$

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COMPONENT MATRICES

$$U \in \mathbb{R}^{m \times m}$$

$$\Sigma \in \mathbb{R}^{m \times n}$$

$$V \in \mathbb{R}^{n \times n}$$

SECTION 07

Statistical result · `math stats`

## Effect of the treatment.

$$\hat{\beta} = 0.42 \pm 0.03$$

95% CI: [0.36, 0.48]

$p < 0.001$  ·  $n = 1,204$

*For every additional unit of exposure, the outcome rises by 0.42 SD — roughly an **8%** shift on the baseline. Effect size is the headline; the  $p$ -value just rules out chance.*

# Seven variants. One contract.

*math · math derivation · math theorem · math compare ·  
math canvas · math matrix · math stats*

*KaTeX renders the equations; Lattice provides the room around  
them.*